

Decision	Reason	Evidence	Alternative Considered
Performed dataset overview (shape, info(), describe())	To understand the structure, size, data types, and statistical characteristics before analysis.	Dataset dimensions, data types, and summary statistics were generated.	Starting visualization directly was rejected because understanding the data is a prerequisite for meaningful analysis.
Checked for missing values	Missing data can affect statistical analysis and visualizations.	Missing value summary and heatmap showed that only travel_agent_id contains missing values.	Ignoring missing values was rejected because missing data should always be assessed before analysis.
Retained missing values in travel_agent_id	Missing values represent bookings made without a travel agent and therefore have business meaning.	Only one column contained missing values, while all other attributes were complete.	Removing these rows was rejected because it would unnecessarily reduce the dataset size and remove valid direct bookings.
Checked duplicate records	Duplicate bookings can bias descriptive statistics and business conclusions.	Duplicate count was computed before preprocessing.	Ignoring duplicates was rejected because duplicates may distort booking counts and trends.
Converted reservation_date to datetime format	Date variables stored as text cannot be used efficiently for time-based analysis.	reservation_date changed from object to datetime64[ns].	Keeping the column as text was rejected because monthly and seasonal analyses require proper datetime values.
Used count plots for categorical variables	Count plots clearly compare booking frequencies across different categories.	Booking status, hotel type, booking season, booking channel, and customer category were visualized.	Pie charts were rejected because several variables contain multiple categories where bar charts are easier to compare.
Used histogram for Average Daily Rate	To study the distribution of room prices and identify skewness or unusual pricing behaviour.	ADR histogram showed most prices concentrated in the mid-price range with a right tail.	A boxplot alone was rejected because it does not clearly show the overall distribution.
Used monthly booking count plot	To identify seasonal booking trends throughout the year.	Monthly booking distribution highlighted peak and low-demand months.	Quarterly aggregation was rejected because monthly analysis provides finer detail.
Used boxplots for outlier detection	Boxplots effectively identify extreme values in numerical variables.	Numerical columns were inspected for potential outliers.	Only using summary statistics was rejected because visual inspection better highlights outliers.
Used correlation heatmap	To identify relationships among numerical variables.	Strong correlations were observed between weekday nights and total stay nights, family booking and children, etc.	Pairwise numerical tables alone were rejected because they are difficult to interpret visually.

Used grouped count plot for hotel type vs booking status	To compare cancellation behaviour between hotel types.	Urban Hotels showed more cancellations in absolute numbers because they also have higher booking volumes.	Separate count plots were rejected because grouped visualization allows direct comparison.
Used Average Daily Rate by booking segment	To compare revenue potential across different booking segments.	Online Travel Agencies and Direct bookings showed higher average ADR.	Comparing only booking counts was rejected because it does not measure revenue contribution.
Based recommendations on analytical evidence	Business recommendations should directly follow observed patterns rather than assumptions.	Recommendations were linked to booking trends, pricing, cancellations, and customer behaviour.	Generic hotel management recommendations were rejected because they were not supported by the analysis.